Curvature, Connections, and Characteristic Classes

A Presentation on Principal Bundles

April 17, 2026

Curvature of the Connection on a Principal Bundle

There is some $F_a \in \Omega^2(M; \operatorname{ad} P)$ such that

$$\pi^* F_a = da + \frac{1}{2}[a \wedge a]$$

Definition 50

The 2-form F_a defined by the equation above is called the curvature form of a.

Example: $\mathbb{C}P^n$

U(1) action on S^{2n+1}

The infinitesimal action of $U(1)$ on S^{2n+1} is given by the vector field

$$v(z) = (iz_0, \ldots, iz_n)$$

There is a unique connection $a \in \Omega^1\left(S^{2n+1}; i\mathbb{R}\right)$ such that $\ker a = v^{\perp}$. Explicitly,

$$a_z(u) = \langle v(z), u\rangle i, \quad \text{where } u \in T_z S^{2n+1}$$

For $\mathbb{C}P^1$

$$\pi^* F_a = da = 2\left(dx_0 \wedge dx_1 + dx_2 \wedge dx_3\right) i$$

In particular, we have

$$F_a\left(\pi_* v_2, \pi_* v_3\right) = \pi^* F_a\left(v_2, v_3\right) = 2i$$

We conclude that $F_a = 2\,\mathrm{vol}_{S^2_{1/2}}\, i$. Notice that

$$\int_{S^2} F_a = 2\,\mathrm{Vol}\left(S^2_{1/2}\right) i = 2\pi i$$

Differential on Horizontal (Basic) Forms

PROPOSITION 5.1

If φ is a pseudotensorial (G-equivariant) r-form on P of type (ρ, V), then

- (a) The form φh defined by $(\varphi h)(X_1, \ldots, X_r) = \varphi(hX_1, \ldots, hX_r)$, $X_i \in T_u(P)$, is a tensorial (basic) form of type (ρ, V);
- (b) $d\varphi$ is a pseudotensorial $(r+1)$-form of type (ρ, V);
- (c) The $(r+1)$-form $D\varphi$ defined by $D\varphi = (d\varphi)h$ is a tensorial form of type (ρ, V).

Covariant Derivative Examples

1) Action on $\Omega^1_{\text{bas}}(P;\mathfrak{g})^G$

For $\varphi \in \Omega^1_{\text{bas}}(P;\mathfrak{g})^G$, a - connection form:

$$D\varphi(X,Y) = d\varphi(X,Y) + \frac{1}{2}[\varphi(X), a(Y)] + \frac{1}{2}[a(X), \varphi(Y)]$$

2) Relation to Curvature

$$Da = \pi^* F_a$$

Differentiating Sequence

$$
\begin{tikzcd}
{} \arrow[r, "D"] & \Omega^k_{\mathrm{bas}}(P;\mathfrak{g})^G \arrow[r, "D"] & \Omega^{k+1}_{\mathrm{bas}}(P;\mathfrak{g})^G \arrow[r, "D"] & {} \\
{} \arrow[r, "d_\nabla"] & \Omega^k(M;\mathrm{ad}P) \arrow[r, "d_\nabla"] \arrow[u, "\pi^*"] \arrow[d] & \Omega^{k+1}(M;\mathrm{ad}P) \arrow[r, "d_\nabla"] \arrow[u, "\pi^*"] \arrow[d] & {} \\
{} \arrow[r, "d_\nabla"] & \Omega^k(M;P\times_\rho V) \arrow[r, "d_\nabla"] & \Omega^{k+1}(M;P\times_\rho V) \arrow[r, "d_\nabla"] & {}
\end{tikzcd}
$$

Relation to Curvatures on Vector Bundles

Proposition 52

Let a be a connection on a principal bundle P and let V be a G-representation. We can think of F_a as a 2-form with values in $\mathrm{End}\,(P \times_\rho V)$. With these identifications in mind, the curvature of the induced connection ∇ on $P \times_\rho V$ equals F_a.

Bianchi Identity

The exterior covariant derivative of the curvature vanishes.

$$D\pi^* F_a = 0$$
$$d_\nabla F_a = 0$$

Gauge Group

Definition

For a principal bundle $P \to M$, the gauge group is the automorphism group

$$\mathcal{G}(P) := \{\psi : P \to P \mid \pi \circ \psi = \pi, \psi(pg) = \psi(p)g \quad \forall p \in P, g \in G\}$$

Since $\psi(p)$ lies in the same fiber as p, we can write $\psi(p) = p\hat{f}(p)$ for some $\hat{f} : P \to G$.

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Since $\psi(p)$ lies in the same fiber as p, we can write $\psi(p) = p\hat{f}(p)$ for some $\hat{f} : P \to G$. The equivariancy of ψ is equivalent to $\hat{f}(pg) = g^{-1}\hat{f}(p)g$. This gives a map from the gauge group to sections of the adjoint bundle:

$$\mathcal{G}(P) \to \Gamma(\mathrm{Ad}\, P)$$

Action of the Gauge Group

Exercise 59

Fix some non-negative integer k. Prove that $C^k(M; \mathrm{Ad}\, P)$ is a Banach Lie group. Moreover, show that the Lie algebra of this group is $C^k(M; \mathrm{ad}\, P)$.

Action on Connections

The gauge group acts on the space of connections $\mathcal{A}(P)$:

$$\mathcal{A}(P) \times \mathcal{G}(P) \to \mathcal{A}(P), \quad (a, \psi) \mapsto a \cdot \psi := \psi^* a$$

Exercise 62

Let $E := P \times_{G,\rho} V$ be an associated vector bundle. Show that the following:

(i) The representation ρ induces a natural homomorphism of gauge groups $\gamma : \mathcal{G}(P) \to \mathcal{G}(E)$;

(ii) The corresponding infinitesimal map $d\rho_e : \mathfrak{g} \to \operatorname{End} V$ induces a Lie algebra homomorphism $C^k(M; \operatorname{ad} P) \to C^k(M; \operatorname{End} E)$;

(iii) $\nabla^{a \cdot g} = g^{-1} \nabla^a g$. Here $a \in \mathcal{A}(P)$, $g \in \Gamma(\operatorname{Ad} P)$, and ∇^a denotes the connection on E induced by a.

Exercise 63

Show that the infinitesimal action of the gauge group is given by

$$\mathcal{A}(P) \times \Gamma(\operatorname{ad} P) \to \Omega^0(\operatorname{ad} P), \quad (a, \xi) \mapsto -d_a \xi.$$

Levi-Civita Connection

Torsion

The torsion of a connection ∇ is the tensor field

$$T(v, w) := \nabla_v w - \nabla_w v - [v, w]$$

Theorem (Theorem 68)

For any Riemannian manifold (M, g) there is a unique metric, torsion-free connection ∇. This is the Levi-Civita connection.

Classification of U(1) and SU(2) Bundles

First Chern Class c_1

To any principal $\mathrm{U}(1)$-bundle $P \to M$, we can associate a cohomology class. If $f : M \to \mathbb{C}P^\infty$ is a classifying map, then the first Chern class is:

$$c_1(P) := -f^* a \in H^2(M; \mathbb{Z})$$

For a complex line bundle $L \to M$, we define $c_1(L) := c_1(P)$ where P is a $\mathrm{U}(1)$-structure on $\mathrm{Fr}(L)$.

Second Chern Class c_2

Let $P \to M$ be a principal $\mathrm{Sp}(1)$-bundle (recall $\mathrm{Sp}(1) \cong \mathrm{SU}(2)$). If $f : M \to \mathbb{H}P^\infty$ is a classifying map, the second Chern class is:

$$c_2(P) := -f^* b \in H^4(M; \mathbb{Z})$$

Chern-Weil Theory: Invariant Polynomials

Let G be a Lie group with Lie algebra $\mathfrak{g}$. Let $I^k(G)$ be the set of symmetric multilinear mappings $f : \mathfrak{g} \times \cdots \times \mathfrak{g} \to \mathbb{R}$ such that

$$f\left((\operatorname{ad} a)t_1, \ldots, (\operatorname{ad} a)t_k\right) = f\left(t_1, \ldots, t_k\right)$$

for $a \in G$ and $t_1, \ldots, t_k \in \mathfrak{g}$. A multilinear mapping f satisfying the condition above is said to be invariant (by G).
We set the algebra of invariant polynomials as:

$$I(G) = \bigoplus_{k=0}^{\infty} I^k(G)$$

Ad-invariant Homogeneous Polynomials

Let $p : \mathfrak{g} \to \mathbb{C}$ be an ad-invariant homogeneous polynomial of degree d. This means:

- Given a basis $\xi_1, \ldots, \xi_n$ of $\mathfrak{g}$, the expression $p\left(x_1\xi_1 + \cdots + x_n\xi_n\right)$ is a polynomial of degree d in $x_1 \ldots x_n$.
- $p\left(\mathrm{ad}_g\, \xi\right) = p(\xi)$ for all $g \in G$ and all $\xi \in \mathfrak{g}$.
- $p(\lambda\xi) = \lambda^d p(\xi)$ for all $\lambda \in \mathbb{R}$ and all $\xi \in \mathfrak{g}$.

There is an isomorphism of algebras $I(G) \cong P(G)$, where $P(G)$ is the algebra of ad-invariant polynomials.

Applying Polynomials to Curvature

We can take wedge products of forms with values in a vector space:

$$(\omega_1 \otimes v_1) \wedge (\omega_2 \otimes v_2) = \omega_1 \wedge \omega_2 \otimes (v_1 \otimes v_2)$$

For the curvature form $\Omega \in \Omega^2(P, \mathfrak{g})$, we can form powers:

$$\Omega^k = \Omega \wedge \cdots \wedge \Omega \in \Omega^{2k}\left(P, \otimes_{i=1}^k \mathfrak{g}\right)$$

We can then apply an invariant polynomial $f \in I^k(G)$ to Ω^k:

$$f(\Omega) := f(\Omega^k) \in \Omega^{2k}(P)$$

$$f(\Omega)(\xi_1, \ldots, \xi_{2k}) = \frac{1}{(2k)!} \sum_{\sigma \in S_{2k}} \mathrm{sgn}(\sigma) f\left(\Omega\left(\xi_{\sigma(1)}, \xi_{\sigma(2)}\right), \ldots, \Omega\left(\xi_{\sigma(2k-1)}, \xi_{\sigma(\right.\right.$$

Chern-Weil Homomorphism: Main Theorem

Theorem

Let P be a principal fibre bundle over M with group G. Let Ω be the curvature form of a connection on P.

1. *For each $f \in I^k(G)$, the $2k$-form $f(\Omega)$ on P is basic, i.e., it projects to a unique closed $2k$-form $\bar{f}(\Omega)$ on M such that $f(\Omega) = \pi^*(\bar{f}(\Omega))$.*
2. *The de Rham cohomology class $[\bar{f}(\Omega)] \in H^{2k}(M; \mathbb{R})$ is independent of the choice of connection.*
3. *The map $w : I(G) \to H^*(M; \mathbb{R})$ given by $f \mapsto [\bar{f}(\Omega)]$ is an algebra homomorphism, called the Chern-Weil homomorphism.*

Chern Classes via Chern-Weil

Characteristic Polynomial

Choose $\mathfrak{g} = \mathfrak{u}(r)$ and define polynomials $c_1, \ldots, c_r$ by the equality:

$$\det\left(\lambda I + \frac{i}{2\pi}\xi\right) = \lambda^r + c_1(\xi)\lambda^{r-1} + \cdots + c_r(\xi)$$

These c_j are Ad-invariant polynomials of degree j.

Definition (Definition 84)

The *j-th Chern class* of P is the cohomology class

$$c_j(P) := [c_j(F_A)] \in H^{2j}_{dR}(M; \mathbb{R})$$

The *total Chern class* of P is

$$c(P) := 1 + c_1(P) + \cdots + c_r(P) \in H^\bullet(M; \mathbb{R})$$